2
Ceramic Proton Conductors

Vineet K. Gupta and Jerry Y. S. Lin

2.1
Introduction

Perovskite-structured ceramics with mixed protonic–electronic conductivity at high temperature (>600 °C) have attracted considerable attention in recent years as nonporous, hydrogen permselective membranes. One of the major driving forces for this increasing interest is the need for large quantities of pure H_2, produced through conversion of methane or coal at high temperatures [1–3]. These mixed proton-conducting membranes may be incorporated in coal-fired power plants to separate H_2 from syngas (CO + H_2) at 900 °C. The H_2 separation is achieved in a nongalvanic mode, without the need for an external power supply and hence offers economic advantages over existing technologies [4]. Figure 2.1 is a schematic showing the mechanism of mixed proton and electron-conducting ceramic membranes.

For most proton conductors, it is usually not possible to have both high conductivity and thermal stability due to the weakness of hydrogen bonds. But in 1981, Iwahara et al. [5] discovered that certain perovskite-type ceramics can have appreciable protonic and electronic conduction at high temperatures in H_2 atmospheres. These perovskite-type ceramics are based on $SrCeO_3$ in which some trivalent cations are substituted for tetravalent Ce^{4+}. They include $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$, $SrCe_{0.95}Y_{0.05}O_{3-\delta}$, and $SrCe_{0.95}Sc_{0.05}O_{3-\delta}$. Proton conduction was also demonstrated in other doped perovskite-type ceramics such as rare-earth doped $BaCeO_3$ [6–8].

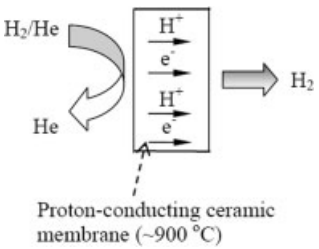


Figure 2.1 Hydrogen transport through a dense mixed protonic–electronic membrane.

Nonporous Inorganic Membranes.
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Nonporous ceramic membranes have advantages of infinite selectivity to H_2 and excellent thermal stability at high temperatures. Mesoporous inorganic membranes (pore diameter 4–5 nm) based on alumina, titania and zirconia are not suitable for high temperature H_2 separation due to their low H_2 permselectivity (governed by the Knudsen mechanism) [9]. Other inorganic membranes that show good H_2 permselectivity at high temperatures are made of amorphous microporous (pore diameter <2 nm) carbon and silica, dense amorphous silica, and microporous crystalline zeolite [10, 11]. The microporous carbon membranes show good selectivity but do not offer the desired H_2 permeance and stability. The sol–gel derived microporous silica membranes offer both good selectivity (selectivity of H_2 to CH_4 is about 10^4) and permeance for H_2. However, the major shortcoming for these silica membranes is their poor hydrothermal stability [11]. Crystalline palladium alloy membranes [10] offer the highest permeability in the temperature range 400–600 °C, but H_2 embrittlement (due to a phase transformation) and sulfur poisoning of the palladium layer are problems with these membranes. The only group of inorganic membranes that can be used at higher temperatures (>800 °C) are the perovskite-structured, proton-conducting ceramic membranes.

Potential applications of these membranes include steam electrolysis [6, 12], H_2 separation from hydrogen containing gases [13, 14], H_2–air fuel cells [15, 16], H_2 sensor devices [17, 18] and membrane reactors for hydrocarbon conversion [19–22]. In the application of methane conversion, proton-conducting membranes have a unique advantage over oxygen-ionic conducting membranes because they operate in a dehydrogenation mode and can avoid overoxidation of CH_4 into CO, CO_2 and H_2O. Figure 2.2 compares the principles of CH_4 conversion for a mixed-protonic conducting membrane and a mixed-oxide conducting membrane. In another application, proton conducting membranes are employed in a reactor where steam-reforming reactions take place on one side and H_2 combustion on the other side of the membrane [23]. The heat of H_2 combustion is exchanged through the membrane to provide heat for the endothermic steam reforming reaction.

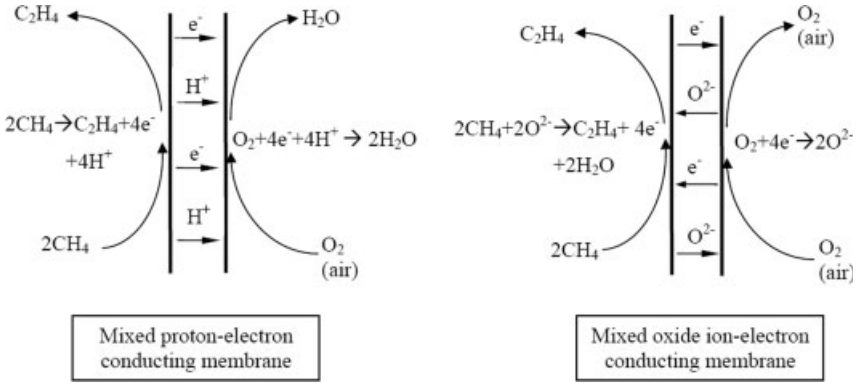


Figure 2.2 Schematic of dehydrogenation reactions of methane illustrating the different functions of mixed proton and oxide ion conducting membranes.

This chapter addresses several aspects of nonporous, proton–electron conducting membranes and presents a concise overview of several studies conducted in these areas. First, the transport properties of several mixed protonic conductors are discussed followed by a presentation of methods to increase the electronic conductivity. New developments in the synthesis route to perovskite powders and thin films are also reviewed. Hydrogen permeation studies of different membranes reported in the literature are summarized, and the effects of the various parameters on the H_2 flux are discussed. A summary of the various mathematical models to describe H_2 flux is presented to elucidate the different approaches adopted and their shortcomings. A concise overview of recent studies related to the stability of the perovskite-type ceramics under CO_2, H_2O, and reducing atmospheres is presented, with discussion of the methods to improve the chemical stability against each gas with minimized negative impacts on other transport properties of the materials. The chapter concludes with some suggestions for future research for mixed proton-conducting membranes.

2.2
General Properties of Perovskite-structured Proton-conducting Ceramic Membranes

2.2.1
Creation of Protonic Carriers

Unlike general ionic conductors, protons are not a part of the nominal structure or stoichiometry of the perovskite-based protonic conductors. The protons are present as foreign species (defects) that remain from synthesis conditions or which are in equilibrium with ambient H_2 and water vapor [24, 25]. Doping by aliovalent cations is essential for the appearance of the protonic conduction in these materials [26]. This is related to the oxygen ion vacancies in the lattice structure generated by doping. In wet gases, the equilibrium of oxygen vacancies of doped $SrCeO_3$ with O_2 partial pressure and water vapor in ambient atmosphere is illustrated by the following equations:

$$V_O^{\bullet\bullet} + \tfrac{1}{2}O_2 \Leftrightarrow O_O^x + 2h^{\bullet} \tag{2.1}$$

$$H_2O + V_O^{\bullet\bullet} + O_O^x \Leftrightarrow 2OH_O^{\bullet} \tag{2.2}$$

These equations and the symbols are in accordance with Kröger–Vink notation [27]. $V_O^{\bullet\bullet}$ is an oxygen vacancy, O_O^x is the lattice oxygen, $h^{\bullet}$ is an electronic hole, $OH_O^{\bullet}$ is a hydroxyl ion, which represents an interstitial proton associating strongly with a neighboring oxygen ion. From the equilibrium of reactions (2.1) and (2.2), the activity of electronic-holes and protons can be described as:

$$[h^{\bullet}] = K_A[V_O^{\bullet\bullet}]^{1/2}(p_{O2})^{1/4} \tag{2.3}$$

$$[OH_O^{\bullet}] = K_B[V_O^{\bullet\bullet}]^{1/2}(p_{H_2O})^{1/2} \quad (2.4)$$

From Eq. (2.3) and (2.4) it is apparent that protonic and p-type electronic conduction depend respectively, on the H_2O partial pressure and O_2 partial pressure.

It is generally accepted that once protons are incorporated into the material they are conducted by hopping between oxygen ions at normal lattice positions at moderate temperatures (<800 °C) [28, 29]. It is also known that the oxide ion migration occurs at high temperatures [30, 31]. The coexistence of these two mechanisms has resulted in speculation about hydroxyl ion migration. Hence, it is possible that protonic conduction occurs by both a hopping mechanism and hydroxyl ion migration to some extent, and that the relative contribution to the overall conductivity will depend on the material composition, operating temperature, and gas composition. Although defect chemistry reactions successfully describe the generation of protons and holes, they do not offer any hint as to why the chemical equilibria are favorable in some perovskites, as opposed to other oxygen ion vacancy rich structures, such as yttria-stabilized zirconia. In the literature, only a few studies suggest a satisfactory explanation for this. It could be possible that these processes are facilitated by the open perovskite structure and the chemical tendency of Ba^{2+} and Sr^{2+} to hydrate [32].

From the quantum molecular dynamics simulations of ABO_3 type ceramics, it was suggested that the degree of covalence between B site cations and the oxygen anions and the degree of hydrogen bonding within the lattice could be the critical factors that influence the conductivity [33–35]. Materials with relatively open crystal structure have greater separation between oxygen anions and a tendency for low B–O covalence. The low covalence results in softer B–O vibrations. As oxygen anions move closer during a vibration, the energy barrier for proton transfer diminishes, facilitating transfer of protons between oxygen ions. It is also expected that a higher degree of hydrogen bonding leads to greater protonic conduction because in this case protons are also in contact with the adjacent oxygen anions, making the proton transfer rapid. However, strong hydrogen bonding is associated with more closely packed structures that have smaller separations between oxygen anions and stiffer B–O vibrations. Therefore, based on quantum mechanical simulations, a compromise between the oxygen–oxygen separation and the stiffness of B–O bonds must be achieved to maximize proton conductivity. From these simulations, the calculated barrier for proton conduction was found to be lower for $BaCeO_3$ than the more closely packed $BaZrO_3$.

2.2.2 Transport Properties

In general the charge carriers in mixed proton–electron conducting materials could be protons, oxygen ions (or oxygen vacancies), hydroxyl ions, electronic-holes and electrons. To determine the nature of the charge carriers and their transport numbers, various defect structure models have been presented [36–38]. These defect models were fitted to the total electrical conductivity data measured

under different controlled atmospheres. In $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$, the total electrical conductivity falls linearly with the square root of the water vapor pressure at constant oxygen partial pressures, indicating the presence of protonic conduction [39]. The transport number of the oxygen ions was determined from measurements of the total electrical conductivity at different partial pressures of O_2. It was found that the total conductivity of some $SrCeO_3$-based protonic conductors depends linearly on the one-fourth power of the oxygen partial pressure [40–43].

The direct evidence for protonic conduction was confirmed primarily by studying the emf of the following gas concentration cell at high temperatures using the specimen ceramics as the electrolyte diaphragm [44–47]:

gas 1, Pt | ceramic disc| Pt, gas 2 (2.5)

Air, O_2, N_2 and H_2 were used as gas 1 or gas 2, in wet or dry states. The relevant ionic transport number is then obtained by comparing the measured emf with the predicted emf from the Nernst equation.

Table 2.1 provides the value of the total electrical conductivity of various perovskite-type proton conductors at 900 °C in dry or wet H_2 atmospheres. It also lists the activation energies of the electrical conduction in the range 800–950 °C. The Arrhenius plots of the conductivity for some of these perovskite-type ceramics are presented in Figure 2.3. These ceramics become almost pure protonic conductors

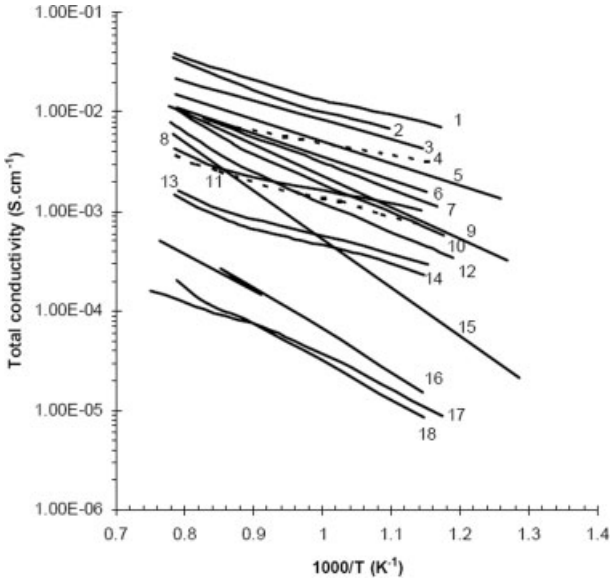


Figure 2.3 Protonic conductivity versus reciprocal of temperature in H_2 atmospheres for various protonic conductors (except Curve 3 where wet H_2 (P_{H2O}= 1.7×10^{-3} atm) is used). 1. $BaCe_{0.95}Y_{0.05}O_{3-\delta}$ [7], 2. $BaCe_{0.90}Nd_{0.10}O_{3-\delta}$ [6], 3. $BaCe_{0.50}Zr_{0.40}Y_{0.10}O_{3-\delta}$ [53], 4. $BaCe_{0.95}Gd_{0.05}O_{3-\delta}$ [7], 5. $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ [48], 6. $BaZr_{0.95}Rh_{0.05}O_{3-\delta}$ [54], 7. $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$ [5], 8. $SrZr_{0.95}Yb_{0.05}O_{3-\delta}$ [50], 9. $SrCe_{0.95}Ho_{0.05}O_{3-\delta}$ [48], 10. $SrCe_{0.95}Y_{0.05}O_{3-\delta}$ [5], 11. $SrZr_{0.95}Y_{0.05}O_{3-\delta}$ [50], 12. $SrCe_{0.95}Sc_{0.05}O_{3-\delta}$ [5], 13. $SrCe_{0.95}Rh_{0.025}Y_{0.025}O_{3-\delta}$ [54], 14. $SrZr_{0.95}In_{0.05}O_{3-\delta}$ [55], 15. $SrCe_{0.95}Eu_{0.05}O_{3-\delta}$ [48], 16. $SrZr_{0.95}Al_{0.05}O_{3-\delta}$ [50], 17. $SrZrO_{3-\delta}$ [50], 18. $SrZr_{0.95}Ga_{0.05}O_{3-\delta}$ [50].

Tab. 2.1 Total electrical conductivity of various protonic conductors.

Material	Total electrical conductivity (S cm^{-1}) at 900 °C	Activation energy kJ mol^{-1} (eV molecule^{-1})	Ref.
$SrCe_{0.95}Yb_{0.05}O_{3-\delta}$	7.2×10^{-3}	59.0 (0.61)	5
$SrCe_{0.90}Y_{0.10}O_{3-\delta}$	6.0×10^{-3}	67.0 (0.69)	5
$SrCe_{0.95}Sc_{0.05}O_{3-\delta}$	3.5×10^{-3}	67.0 (0.69)	5
$SrCe_{0.95}Eu_{0.05}O_{3-\delta}$	2.7×10^{-3}	93.0 (0.96)	48
$SrCe_{0.95}Sm_{0.05}O_{3-\delta}$	2.3×10^{-3}	78.0 (0.81)	48
$SrCe_{0.95}Ho_{0.05}O_{3-\delta}$	6.6×10^{-3}	61.0 (0.63)	48
$SrCe_{0.95}Tm_{0.05}O_{3-\delta}$	1.2×10^{-2}	39.0 (0.40)	48
$SrCe_{0.95}Tb_{0.05}O_{3-\delta}$	7.5×10^{-3}	54.0 (0.56)	49
$SrZr_{0.95}Yb_{0.05}O_{3-\delta}$	2.9×10^{-3}	46.0 (0.48)	50
$SrZr_{0.95}Y_{0.05}O_{3-\delta}$	2.6×10^{-3}	46.0 (0.48)	50
$SrZr_{0.95}In_{0.05}O_{3-\delta}$	9.0×10^{-4}	61.0 (0.63)	50
$SrZr_{0.95}Al_{0.05}O_{3-\delta}$	2.6×10^{-4}	77.0 (0.80)	50
$SrZr_{0.95}Ga_{0.05}O_{3-\delta}$	1.2×10^{-4}	72.0 (0.75)	50
$BaCe_{0.95}Nd_{0.05}O_{3-\delta}$	7.6×10^{-3}	73.0 (0.76)	6
$BaCe_{0.90}Nd_{0.10}O_{3-\delta}$	2.2×10^{-2}	54.0 (0.56)	6
$BaCe_{0.95}Ca_{0.05}O_{3-\delta}$	5.0×10^{-3}	49.0 (0.51)	6
$BaCe_{0.90}La_{0.10}O_{3-\delta}$	1.0×10^{-2}	50.0 (0.52)	6
$BaCe_{0.90}Nd_{0.10}O_{3-\delta}$	3.2×10^{-2}	54.0 (0.56)	31
$BaCe_{0.95}Y_{0.05}O_{3-\delta}$	2.7×10^{-2}	43.0 (0.44)	7
$BaCe_{0.90}Yb_{0.10}O_{3-\delta}$	1.5×10^{-2}	26.0 (0.27)	7
$BaCe_{0.95}Gd_{0.05}O_{3-\delta}$	8.1×10^{-3}	24.0 (0.25)	7
$Ba_{1.03}Ce_{0.80}Er_{0.20}O_{3-\delta}$	1.2×10^{-2}	24.0 (0.25)	52
$BaCe_{0.50}Zr_{0.40}Y_{0.10}O_{3-\delta}$	1.6×10^{-2}	37.0 (0.38)	53
$SrZr_{0.95}Rh_{0.025}Y_{0.025}O_{3-\delta}$	1.1×10^{-3}	38.0 (0.39)	54
$CaZr_{0.95}In_{0.05}O_{3-\delta}$	5.86×10^{-4}	48.0 (0.50)	55
$SrZr_{0.95}In_{0.05}O_{3-\delta}$	8.8×10^{-4}	48.0 (0.50)	55
$BaZr_{0.95}In_{0.05}O_{3-\delta}$	9.7×10^{-4}	50.0 (0.52)	55
$SrTi_{0.90}Ru_{0.10}O_{3-\delta}$	1.3×10^{-1}	45.0 (0.47)	56

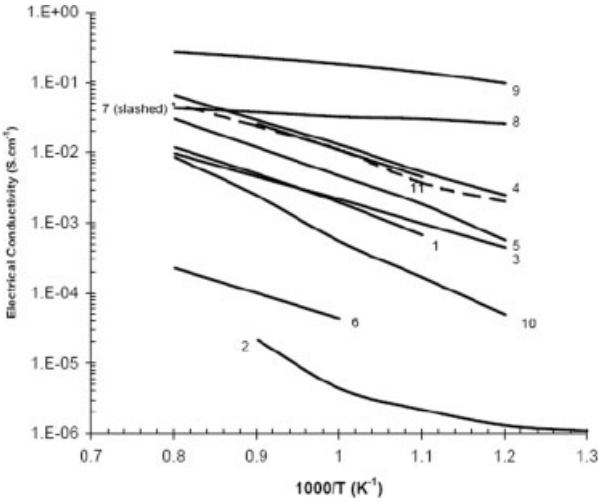


Figure 2.4 Total electrical conductivity versus reciprocal of temperature for various protonic conductors (dry N_2 if not specified).
1. $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$ [5], 2. $SrCe_{0.95}Tb_{0.05}O_{3-\delta}$ [48], 3. $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ [54], 4. $BaCe_{0.90}Nd_{0.10}O_{3-\delta}$ [6] (wet air), 5. $BaCe_{0.95}Nd_{0.05}O_{3-\delta}$ [6] (wet air), 6. $BaCeO_{3-\delta}$ [5] (air), 7. $SrZr_{0.95}Al_{0.05}O_{3-\delta}$ [50] (wet air), 8. $SrCe_{0.90}Mn_{0.10}O_{3-\delta}$ [50] (air), 9. $SrTi_{0.90}Ru_{0.10}O_{3-\delta}$ [56] (wet air), 10. $La_{0.95}Ca_{0.05}ScO_{3-\delta}$ [58], 11. $SrCe_{0.95}Y_{0.05}O_{3-\delta}$ [59] (air).

in H_2 atmospheres, as determined from the emf measurements of the gas concentration cells, and the total electrical conductivity corresponds to the protonic conductivity of these materials, except for $SrTi_{0.90}Ru_{0.10}O_{3-\delta}$ where oxygen ions primarily contribute to the total electrical conductivity. Among Sr-based perovskite-type ceramics, $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ shows the highest protonic conductivity of 1.2×10^{-2} S cm^{-1} while $BaCe_{0.90}Nd_{0.10}O_{3-\delta}$ shows the highest protonic conductivity of 3.2×10^{-2} S cm^{-1} among Ba-based perovskite-type ceramics. The conductivities of zirconate-based ceramics are almost two orders of magnitude lower than those of the cerates, but they are superior with respect to chemical resistance and mechanical strength.

Figure 2.4 shows Arrhenius plots of the total electrical conductivity for some of these perovskite-type ceramics in hydrogen free atmospheres. The protonic conduction is negligible here due to the absence of H_2. In this case, only the electronic, and, in some cases, the oxygen ionic conductivity contribute to the total conductivity. It should be noted from Fig. 2.4 that the total electrical conductivities of $SrTi_{0.90}Ru_{0.10}O_{3-\delta}$ in wet air and $SrCe_{0.90}Mn_{0.10}O_{3-\delta}$ in air are very high. However, the nature of the electrical conductivity is largely electronic and, therefore, these materials are not very attractive for membrane applications because of their low protonic conductivity.

A summary of the various studies determining the transport number of various charge carriers in mixed protonic conductors using different gas-concentration cells is presented in Table 2.2. In $BaCe_{0.95}Y_{0.05}O_{3-\delta}$, and $BaCe_{0.90}Nd_{0.10}O_{3-\delta}$ the nature of the dominant ionic conduction gradually changes from protonic to oxide

Tab. 2.2 Summary of transport numbers of various mixed protonic–electronic conductors in various gas concentration cells.

Electrolyte	Ref.	Gas 1/Gas 2	Total conductivity S cm^{-1}		Transference number at 600 °C			Transference number at 800 °C		
			600 °C	800 °C	O^{2-}	H^+	e^-	O^{2-}	H^+	e^-
$BaCe_{0.90}Nd_{0.10}O_{3-\delta}$	60	H_2/21% O_2	1.5×10^{-2}	4.1×10^{-2}	0.23	0.72	0.05	0.65	0.26	0.09
$BaCe_{0.95}Y_{0.05}O_{3-\delta}$	61	O_2+H_2O	6.0×10^{-3}	2.2×10^{-2}	0.33	0.26	0.41	0.37	0.08	0.55
$BaCe_{0.95}Y_{0.05}O_{3-\delta}$	61	H_2+H_2O	4.4×10^{-3}	1.0×10^{-2}	0.05	0.79	0.16	0.16	0.63	0.21
$SrCe_{0.95}Y_{0.05}O_3$	59	100% O_2/4% H_2	2.9×10^{-3}	1.8×10^{-2}	—	0.74	0.26	—	0.71	0.29
$SrCe_{0.95}Y_{0.05}O_{3-\delta}$	59	4% H_2/0.488%H_2	2.9×10^{-3}	1.2×10^{-2}	—	0.90	0.10	—	0.86	0.14
$SrCe_{0.95}Yb_{0.05}O_{3-\delta}$	19	1% H_2/21% O_2	—	5.8×10^{-3}	—	—	—	—	0.22	0.78

ionic conduction as the temperature is increased from 500 to 800 °C in an atmosphere of O_2/H_2O. In these materials, protonic conduction remains dominant in H_2/H_2O atmospheres throughout the entire temperature range (600–800 °C) in spite of the measurable oxygen ionic transference number. Due to oxygen ion conduction in the presence of H_2O, use of $BaCeO_3$-based perovskites for H_2 separation from industrial gas mixtures such as syngas from power plants that usually contain water vapor can cause oxidation of the feed. Consequently, $SrCeO_3$-based materials may be more suitable in these applications. As can be seen in Table 2.2, doped samples of $SrCeO_3$ are essentially protonic conductors with negligible transfer numbers for oxygen ions [19, 59]. According to Bonanos [62] and Knight and Bonanos [63], this could be due to the more distorted orthorhombic structure of $SrCeO_3$ as compared to that of $BaCeO_3$ which inhibits oxide ion conduction. In H_2-containing atmospheres, both Yb- and Y-doped $SrCeO_3$ have very low electronic transference numbers. For a given value of the total conductivity, the ambipolar diffusion flux density reaches a maximum when the ionic transference number equals the electronic transference number [64, 65]. This implies that relatively low electronic conductivity limits the H_2 flux in these perovskite-type ceramics in H_2 containing atmospheres.

2.2.3
Electronic Conductivity and Its Improvement

As discussed in the previous section, one effective way to improve the H_2 permeability of $SrCeO_3$-based proton-conducting ceramic membranes is to improve their electronic conductivity. Different aliovalent dopants have been studied for their effect on the electronic conductivity. Qi and Lin [66] found that the electronic conductivity is inversely related to the ionization potentials of Ce^{4+} and dopants. Table 2.3 lists experimentally determined electronic conductivities of $SrCeO_3$-based materials in pure O_2 with the ionization potentials [67] of their dopant ions. As shown, the electronic conductivity increases with decreasing ionization potential of the dopants. Among $SrCeO_3$-based materials, Tm-doped $SrCeO_3$ has the

Table 2.3 Electronic conductivity of $SrCeO_{3-\delta}$ (SC), $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$ (SCYb), $SrCe_{0.95}Tb_{0.05}O_{3-\delta}$ (SCTb) and $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ (SCTm) and their ionization potentials.

Material	Charge transfer equation	Ionization potential of the doping ions (eV)	Electronic conductivity (S cm^{-1})
SCTb	$Tb^{3+} + h \rightarrow Tb^{4+}$	39.8	1.21×10^{-5}
SC	$Ce^{3+} + h \rightarrow Ce^{4+}$	36.8	2.82×10^{-4}
SCYb	$Yb^{2+} + h \rightarrow Yb^{3+}$	25.0	8.50×10^{-3}
SCTm	$Tm^{2+} + h \rightarrow Tm^{3+}$	23.7	1.21×10^{-2}

lowest ionization potential and shows the highest electronic conductivity of 1.21×10^{-2} S cm^{-1} at 800 °C. This can be explained by the hopping mechanism of semiconductors [68]. According to the hopping mechanism, the electronic conduction results from the transfer of electrons (or electronic holes) between two neighboring ions of different valence. Hence, the lower ionization potential of dopant ion would make the electronic conduction faster.

The other approach to increase the electronic conductivity of these perovskite-based membranes is to add a metal phase (10–40 vol %). The metal phases studied include palladium, niobium, tantalum, vanadium and zirconium or their binary mixtures [69–72]. In order to minimize the stress at internal interfaces that can lead to the formation of dislocations and initiation of cracks, the ceramic support materials were chosen so as to be lattice matched to the metals and metal alloys [73].

2.3 Synthesis of Proton-conducting Ceramic Membranes

2.3.1 Synthesis of Powders

$BaCeO_3$- and $SrCeO_3$-based powders have been synthesized mainly by the conventional solid-state reaction method: $BaCO_3 + CeO_2 \rightarrow BaCeO_3 + CO_2$. In this route, powders need to be ball-milled and calcined for prolonged periods of time, which can lead to a large particle size, high impurity content, nonuniform particle size distribution, and chemical inhomogenity. In general, nanosized ceramic powders with good homogeneity can be prepared from wet chemical methods such as sol–gel [74–76], spray-dry processing of chemically modified alkoxides [77, 78], co-precipitation of mixed metal oxalates [79–83], hydrothermal synthesis [84, 85] and combustion of polymerized complexes [86–89].

Among the various wet chemical methods, the complex precursor synthesis based on the method of Pechini [89] has attracted considerable attention. In this method, the desired cations are solvated in a solution using a chelating agent. Chelating agents such as citric acid or oxalic acid containing carboxylic groups, or glycine and ethylenediaminetetraacetic acid (EDTA) containing carboxylate and aliphatic amine groups were often used. It is generally believed that the metal ions are chelated with citric acid through the –OH and –COOH groups and serve as bridges between the citric acid molecules to form polymeric molecules [90]. The general idea is to achieve a uniform mixing of cations at the molecular level throughout the polymeric resin. The polymeric resin is self-ignited in air, and the resultant powder is calcined to obtain a single phase, chemically homogenous powder of perovskite structure. Liu et al. [88] determined the optimum ratio of chelating agent to the metal cations required to obtain the homogeneous resin. The minimum calcination temperatures required to obtain the single phase $SrCeO_3$ [88] and $BaCeO_3$ [91] powder are 950 and 900 °C, respectively, while single

phase $SrTiO_3$ [87] powders were obtained at 400 °C. Higher calcination temperature requirements to obtain single phase Sr and Ba cerates are due to the formation of carbonates on burning their polymeric resins, unlike the Sr–Ti polymeric resin [87] which does not dissociate into strontium carbonate [88]. The calcination temperature needed to obtain single phase Ba and Sr cerates via the solid-state method is 1200 °C, almost 300 °C higher than that of the complex precursor method.

2.3.2 Effect of Synthesis Conditions on Membrane Performance

The procedure for preparing perovskite-type membranes usually consists of three steps: powder synthesis, shaping, and sintering. In general, the synthesis routes should be assessed according to their ability to produce a single-phase perovskite structure, maintain correct stoichiometry, and exhibit high surface area for high sinterability of the powders, in addition to the practicalities surrounding each technique. The shaping process is also a factor in the membrane performance. For example, the membrane prepared by isostatic pressing is denser than that prepared by extruding [92]. The sintering profile (atmosphere, heating/cooling rate, highest sintering temperature, and dwell time) can obviously affect the membrane microstructure.

Whilst it is widely appreciated that the synthesis route affects the powder properties and subsequently the final properties of a ceramic material, there appears to be no overall consensus in the literature regarding which synthesis method gives optimum properties for membrane materials. Li et al. [93] compared the permeation flux of perovskite-structured $La_{0.20}Sr_{0.80}Co_{0.20}Fe_{0.80}O_{3-\delta}$ (LSCF) membranes prepared by three different methods, namely solid-state, EDTA pyrolysis, and citric acid pyrolysis. Although the O_2 permeation flux of the LSCF membranes prepared by the three methods were very similar, the solid-state reaction method was found to be preferable as it is easily controllable and effective. Tan et al. [94] examined $Ba_{0.50}Sr_{0.50}Co_{0.80}Fe_{0.20}O_{3-\delta}$ (BSCF) membranes in terms of their microstructure and O_2 permeation flux for membranes prepared by three different methods under the same sintering profile. The permeation flux of the membrane prepared by the solid-state method was reported to have higher O_2 flux than those prepared by the citrate and EDTA methods. It was observed that the membranes prepared by the solid-state method have the largest grain sizes, and hence the lowest grain boundary area. Because grain boundaries hinder oxygen ion transport, fewer grain boundaries would favor higher oxygen flux. Kharton et al. [95] obtained similar results for lanthanum cobaltite membranes. Qi et al. [96] concluded that the deviation of composition from the desired stoichiometry, which resulted from different synthesis methods, was the major reason for the discrepancies in the oxygen permeation data.

Dionysiou et al. [97] controlled the porosity of $SrCe_{0.95}Tb_{0.05}O_{3-\delta}$ membranes by varying the intermediate calcination temperature of the green powder and the final sintering temperature of the membrane disks. The membranes made of

powders calcined at higher temperature tend to be porous and gas-permeable. The porous membranes could be useful as substrates to allow faster permeation of H_2. Mukasyan et al. [98] determined the optimum temperature required to reach the final density of the membranes. Two main factors that influence the final membrane density are the green sample density and the powder sinterabilty. The former is higher for powders with lower surface area, while the latter is higher for those with higher surface area.

2.3.3 Preparation of Thin Films

Perovskite-structured membranes, in the form of thin films supported on porous ceramic or metal substrates, have been studied extensively in the past decade. Thin films offer several advantages including reduced material cost, improved mechanical strength and possibly higher H_2 flux. Chemical vapor deposition (CVD) [99], electrochemical vapor deposition (EVD) [100] and sputtering [101] represent typical methods. However, dense films have been difficult to obtain by these methods. It was found that the continuity and gas-tightness of the deposited films were very sensitive to the morphologies and pore size of substrates.

The deposition of perovskite-type ceramic films from solution or a slurry of powders has also been carried out in many studies [102]. In these processes, it is essential to prevent the impregnation of the slurry into the substrate pores. One of the challenges here is to optimize the slurry concentration so that a crack-free film of the desired thickness can be obtained. In one of the studies, Itoh et al. [103] prepared Ca(Ti, Fe)O_3 thin films by spin-coating Ca(Ti, Fe)O_3 powders dispersed in isopropanol onto $CaTiO_3$ porous substrates. $CaTiO_3$ was chosen as the porous substrate material because it is less sinterable and has a similar thermal expansion coefficient as Ca(Ti, Fe)O_3. Visco et al. [104, 105] developed an inexpensive and scalable method of preparation of crack-free electrolyte thin films through systematic elucidation of the shrinkage profiles. The NiO-YSZ substrates were coated with colloidal suspensions of yttria-stabilized zirconia (YSZ) powders prepared by glycine-nitrate methods. Dense YSZ thin films (4–10 μm) were obtained after treatment in air at 1673 K for 1 h. Eschenbaum et al. [106] prepared $SrZr_{0.95}Yb_{0.05}O_{3-\delta}$ thin films on nonporous silicon substrates by depositing a precursor formed via sol–gel methods; treating the composites in air at 873 K in order to remove organic residues, and then sintering at 973–1273 K. Kosaki and Anderson [107] prepared $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$ thin films using a polymeric precursor method. The dense films (~1 μm) were obtained on silica and alumina substrates by a spin-coating technique. Although the films obtained were found to be crack-free, as determined by scanning electron microscopy, the gas tightness of the films was not confirmed by helium permeation.

Recently, Hamakawa et al. [108] spin-coated a $SrCe_{0.95}Yb_{0.50}O_{3-\delta}$ slurry onto a porous $SrCeO_3$ support to obtain thin films of thickness down to 2 μm. A scanning electron micrograph of an asymmetric membrane is shown in Fig. 2.5. However, the synthesis process is rather laborious, and the quality of the film is very

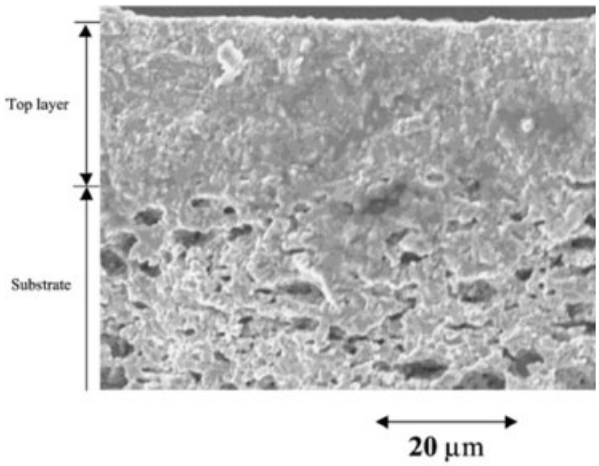


Figure 2.5 Scanning electron micrograph of an asymmetric $SrZr_{0.95}Yb_{0.05}O_{3-\delta}$ membrane [108].

sensitive to the synthesis conditions. Xia and Liu [109, 110] introduced a simple and cost-effective method of pressing dry powders to fabricate the thin, dense ceramic electrolyte layers onto porous substrates of different electrode materials. Cheng et al. [111] adopted the dry pressing method to fabricate thin dense films of $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ onto the porous substrate of the same composition for membrane applications. The smaller size powders are used to make the dense top layer while the larger particle size powders produced porous substrates. The film thickness was varied from 150–800 μm by controlling the amount of powder used for the top layer, which determines the top-layer thickness. Much thinner membranes (<10 μm) cannot be prepared by this method as it is difficult to distribute uniformly a smaller amount of the powder over the supports.

2.4 Hydrogen Permeation

2.4.1 The H_2 Permeation Set-up and Sealing System

The H_2 permeation flux was measured in a nongalvanic mode by imposing a partial pressure gradient of H_2 across the membrane [66, 108, 111, 112]. The general schematic of the high temperature (600–1000 °C) H_2 permeation measurement set-up is shown in Fig. 2.6 [66]. The permeation system consists of three dense alumina (quartz) tubes of different sizes placed coaxially inside a tubular furnace. Disk-shaped or tubular membranes were placed on the top of the ceramic tube. One of the challenges in measuring high temperature hydrogen flux is sealing the membrane. Ceramic-glass composite materials were developed for sealing proton-conducting dense $SrCeO_3$ ceramic membranes [113]. The sealant was com-

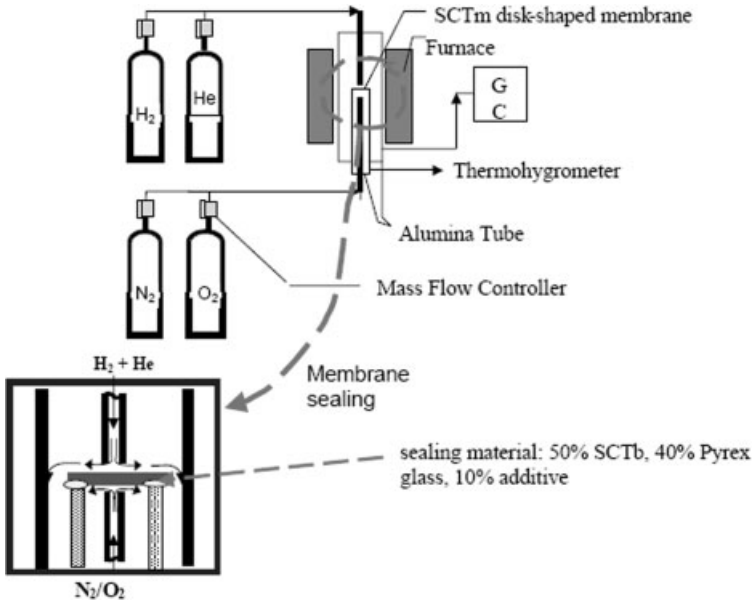


Figure 2.6 Schematic of a high temperature H_2 permeation system [65].

Tab. 2.4 Comparison of hydrogen flux through various nonporous mixed proton conducting membranes at 900 °C.

Membrane	Ref.	Upstream	Downstream	Thickness (mm)	H_2 Flux (mol cm^{-2} s^{-1})
$SrCe_{0.95}Tm_{0.05}O_{3-\delta}$	66	10% He/H_2	20% O_2/N_2	1.6	2.9×10^{-8}
$SrCe_{0.95}Tm_{0.05}O_{3-\delta}$	111	10% He/H_2	20% O_2/N_2	0.15	9.6×10^{-8}
$SrCe_{0.95}Sm_{0.05}O_{3-\delta}$	112	100% H_2	He	1.7	1.1×10^{-9}
$SrCe_{0.95}Eu_{0.05}O_{3-\delta}$	112	100% H_2	He	1.7	3.3×10^{-9}
$SrCe_{0.95}Yb_{0.05}O_{3-\delta}$	108	20% He/H_2	N_2	1.0	9.2×10^{-9}
$SrCe_{0.95}Yb_{0.05}O_{3-\delta}$	108	20% He/H_2	N_2	2.0×10^{-3}	5.2×10^{-7}
$SrCe_{0.95}Y_{0.05}O_{3-\delta}$	59[a]	4% H_2/Ar	Ar	1.6	2.1×10^{-8}
$BaCe_{0.80}Yb_{0.20}O_{3-\delta}$	119[b]	4% H_2/N_2	Ar	1.6	3.6×10^{-8}
$SrCe_{0.95}Yb_{0.05}O_{3-\delta}$	120[b]	1% H_2/Ar	20% O_2/Ar	1.6	1.7×10^{-8}
$BaCe_{0.80}Yb_{0.20}O_{3-\delta}$/ 40% metal	121	4% H_2/He	100 ppm H_2/N_2	0.5	1.3×10^{-7}

a H_2 flux is measured by passing current using Pt electrodes.
b H_2 flux is measured in short-circuited mode using two Pt electrodes.

posed of 45% $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ powder, 50% Pyrex glass, and 5% $NaAlO_2$. Composite-ceramic sealants have advantages over traditional glass or ceramic seals [114–118]. Their properties, such as thermal expansion, wettability to ceramic support and membrane disk, and chemical inertness, can be tailored to match the membrane and the support tube by adjusting sealant constituents. Gas-tight seals between membrane disks and ceramic tube supports divide the system into two chambers. A gas mixture of H_2/He was fed to the upstream side of the permeation system and the downstream side was swept by $H_2/O_2/N_2$. A gas chromatograph (Perkin-Elmer) with a molecule sieve column (1/80D, 10 ft, 801100 mesh, from Alltech) was used to check the seals for leaks by detecting N_2. The H_2 permeation flux was calculated from the rate of water produced at the downstream side as the permeated H_2 reacted with O_2 to produce water. A thermohygrometer (Cole-Parmer) was used to measure the change in the water vapor concentration in the sweep gas that was used to measure the hydrogen flux at steady state.

H_2 flux data in the galvanic mode [59], in which the external current is passed through the electrodes and electrolyte, and in a short-circuited mode [119, 120], was also included. Table 2.4 lists the steady state H_2 permeation flux data of various nonporous mixed protonic conducting membranes. A dual phase membrane of $BaCe_{0.80}Yb_{0.20}O_{3-\delta}$/40% metal has higher H_2 permeability than other membranes listed in Table 2.4 due to the obvious reason of high electronic conductivity of the metal phase. In order to be economically viable, the membrane should provide a H_2 flux of 10 cm^3(STP) min^{-1} cm^{-2} (7.0×10^{-7} mol cm^{-2} s^{-1}) with a pressure drop of 1 atm across the membrane. As can be seen from Table 2.4, this target could be achieved by single phase mixed conducting membranes if their thickness could be reduced to less than 5.0 μm.

2.4.2 Effects of Process Variables on H_2 Flux

2.4.2.1 Effect of Feed and Sweep Side Gas Concentrations

Cheng et al. [111] studied the effect of downstream O_2 partial pressure on the H_2 flux at 900 °C for $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ (SCTm) membranes by keeping the feed gas partial pressure constant. The H_2 flux increases with increasing downstream O_2 partial pressure. It was also found that the H_2 flux increases on increasing the upstream H_2 partial pressure while keeping the sweep side gas concentration constant [111]. These results can be explained by considering the effect of the partial pressure of H_2 and O_2 on the total driving force for H_2 transport and on the conductivity of electrons, holes, and protons in the membrane. Increasing the upstream H_2 partial pressure increases the net driving force for H_2 permeation. It also increases the concentration of protons and electrons on the feed side, hence increasing the overall protonic and n-type electronic conductivity in the membrane. Increasing the O_2 partial pressure on the downstream side of the membrane increases the hole concentration, thereby increasing the overall electrical conductivity of the membrane. In addition, the higher downstream O_2 partial pressure corresponds to a lower H_2 equilibrium partial pressure creating a higher

driving force for H_2 permeation. The same dependence of H_2 permeation flux on the sweep side O_2 partial pressure was observed in a $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$ membrane [120].

2.4.2.2 Effect of Membrane Thickness

Hamakawa et al. [108] and Cheng et al. [111] obtained hydrogen flux data by varying membrane thickness. It was found that the H_2 permeation flux increases with decreasing membrane thickness. The H_2 permeation flux of 2.0 μm thick $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$ (SCYb) membranes was reported to be 5.2×10^{-7} mol cm^{-2} s^{-1}, 500 times greater than for 1.0 mm disks of identical composition [108]. It is important to investigate whether the flux is limited by the bulk diffusion or by the dissociation reaction at the surface. In general, H_2 flux and thickness are plotted on log–log plots. A slope of –1 in such plots implies bulk diffusion limited permeation, and a slope of 0 indicates surface reaction kinetic limited permeation, as described by Chen et al. [122]. A rough estimate of the critical thickness, at which the transition from bulk diffusion to surface reaction kinetics occurs, can be estimated from the ratio of relative rates of bulk diffusion of ions in the membrane and the surface dissociation reactions.

The critical thickness for a $La_{0.30}Sr_{0.70}CoO_{3-\delta}$ (LSCO) membrane, which is a mixed-oxide ion conductor, was determined to be 80 μm at 1000 °C [122]. The ambipolar conductivity of $SrCeO_3$ based material is about 10^{-3} S cm^{-1}, which is lower than the oxide ionic conductivity of the LSCO membrane. Also, the dissociation of H_2 at the surface of the membrane requires less energy than dissociation of O_2 because the H–H bond energy is much lower than the O–O bond energy. This indicates that the critical thickness for $SrCeO_3$-based conductors should be less than 80 μm. It was found that that the H_2 permeation is limited by diffusion, even for 2 μm thin $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$ membranes. Cheng et al. [111] reported a

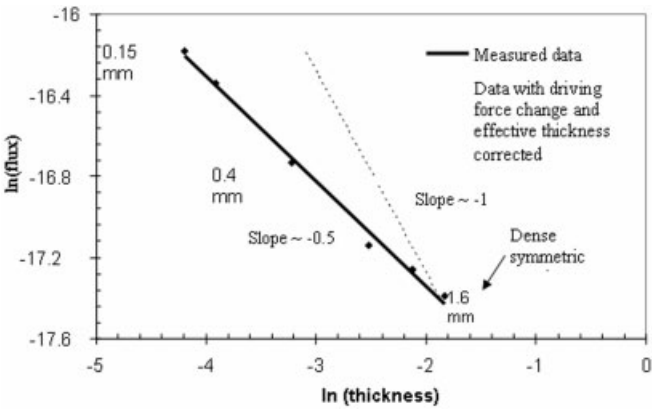


Figure 2.7 Effect of membrane thickness on the H_2 permeation flux of $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ membranes [111].

slope of about –0.5 on the log–log plot of hydrogen flux versus thickness in the thickness range 1.6 mm to 150 μm for $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ membranes. The deviation of the slope from –1 could be due to the permeation of hydrogen through the dense part of the porous support made of SCTm. It could also be due to the fact that the driving force for H_2 permeation is smaller for thinner membranes because of higher downstream H_2 partial pressures as a result of larger permeation flux. If the change in the driving force and the difference between effective and estimated thickness are considered, one will obtain a straight line with a slope closer to –1, as indicated in Fig. 2.7 by the dashed straight line passing through the data points of the 1.6 mm thick symmetric SCTm membrane.

2.4.2.3 **Effect of Temperature**

Figure 2.8 compares Arrhenius plots of ln[H_2 flux] vs. $1/T$ for various mixed proton conducting ceramic membranes. For $SrCe_{0.95}Eu_{0.05}O_{3-\delta}$ (SCEu) and $SrCe_{0.95}Sm_{0.05}O_{3-\delta}$ (SCSm) membranes, the activation energies of the H_2 permeation was determined to be 80 and 85 kJ mol^{-1}, respectively, in the temperature range 600–850 °C [112]. For $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ (SCTm) membranes, the activation energy of H_2 permeation was determined to be 116 kJ mol^{-1} in the lower temperature region (≤700 °C) but decreased to 20 kJ mol^{-1} in the higher temperature region (700–950 °C) [111]. This change in activation energy indicates that the H_2 permeation was limited by the electronic conduction between 600 and 700 °C and by mixed electronic-protonic conduction between 750 and 950 °C. This difference in the activation energy behavior between SCTm and SCEu is due to the different environments in which these membranes are placed. Both membranes were exposed to 10% H_2/He in the feed side, but N_2 was used as a sweep gas in the case of SCEu, and a 20% O_2/N_2 was used as a sweep gas with SCTm.

Hamakawa et al. [120] reported that for SCYb membranes at 900 °C, when N_2 was used as the sweep gas, the electronic conductivity value was much lower than

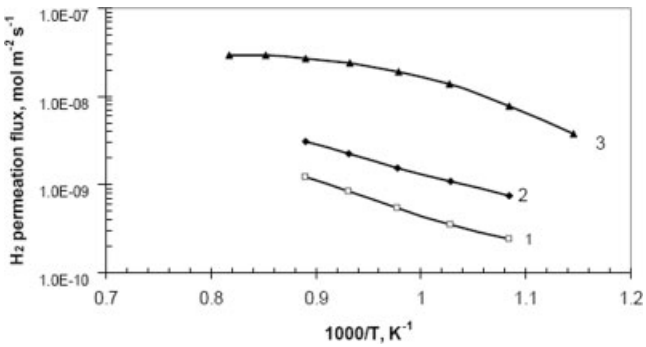


Figure 2.8 Arrhenius plots of hydrogen flux for various membranes. 1. $SrCe_{0.95}Sm_{0.05}O_{3-\delta}$ [112], 2. $SrCe_{0.95}Eu_{0.05}O_{3-\delta}$ [112], 3. $SrCe_{0.95}Tm_{0.05}O_{3-\delta}$ [111].

the protonic conductivity, while protonic and electronic conductivity values were almost equal when 20% O_2/N_2 was used as the sweep. Hence, in SCEu, membrane electronic conductivity values remained low as compared to the protonic conductivity, even at higher temperatures, while in SCTm in the higher temperature ranges the electronic conductivity became comparable to the protonic conductivity. One implication of this result is that one way to increase the H_2 permeation rate under asymmetric conditions is by improving the protonic conductivity, while maintaining the same level of electronic conductivity. This will increase the activation energy of mixed conduction so the H_2 flux will increase more with increasing temperature than the H_2 flux of SCTm.

2.4.3
Mathematical Models for Hydrogen Permeation

Theoretical models for H_2 permeation through proton-conducting ceramic membranes are of two-fold importance. First, these models help us to understand the underlying transport phenomena, and second, they are useful in designing and optimizing processes utilizing these membranes. In all of the studies that dealt with the modeling of H_2 flux, the starting point is the Wagner equation [123,124]:

$$J_{H_2} = \frac{1}{L}\left\{\frac{RT}{8F^2}\int_{P_{O2}}^{P_{O2}''}\sigma_{tot}t_{H^+}t_{O^{2-}}\,d\ln P_{O2} + \frac{RT}{4F^2}\int_{P_{H2}}^{P_{H2}''}\sigma_{tot}t_{H^+}(t_{O^{2-}}+t_e+t_h)\,d\ln P_{H2}\right\} \quad (2.6)$$

Here the charge carriers include protons (H^+), electrons (e^-), electron-holes (h) and oxygen ion vacancies ($V_O^{\bullet\bullet}$). From experimental evidence, it is known that the first term inside the parentheses in Eq. (2.6) can be neglected for $SrCeO_3$-based materials due to the negligible oxygen ion transport number. Assuming a negligible oxygen ion transport number, one finds:

$$J_{H_2} = \frac{1}{4L}\frac{RT}{F^2}\int_{P_{H2}}^{P_{H2}''}\sigma_{tot}t_{H^+}(t_e+t_h)\,d\ln P_{H2} \quad (2.7)$$

Here, it is assumed that the H_2 dissociation rate at the surface of the membranes is much higher than the bulk diffusivity of the charge carriers. The effect of external mass transfer is also neglected.

It is clear from Eq. (2.7) that an accurate prediction of H_2 flux is possible only when the functional dependences of the transport numbers of various charge carriers are known in the H_2 and O_2 partial pressure. However, in some studies, simplifying assumptions were made to obtain an analytical expression for H_2 flux. Norby and Larring [125] obtained an analytical expression for the hydrogen flux by assuming that the electrons were the dominant charge carriers rather than the protons. In other words, the electronic transport number was assumed to be close to one. The protonic conductivity, remaining in the integral of Eq. (2.7), was corre-

lated to the partial pressure of H_2 from the defect equilibrium relations. Qi et al. [126] developed an analytical model for a general case where one side of the membrane is exposed to H_2 and the other side is exposed to O_2. They integrated Eq. (2.7) by dividing the membrane into two regions based on the nature of the dominant charge carrier for electronic transport. Electronic conduction due to electrons (n-type) was considered dominant in the membrane side exposed to H_2, while the electronic conduction due to holes (p-type) was considered dominant in the membrane side exposed to oxygen.

A schematic of the concentration profiles of various charge carriers within membranes is shown in Fig. 2.9. The protonic and electronic conductivity are presented as empirical functions of H_2 partial pressure to substitute into Eq. (2.7). The final analytical expression for the H_2 flux contained five parameters that included proton, electron and electron-hole conductivity, and their concentrations at a reference state of $P_{H_2} = 1$ atm. The model could not be verified due to the lack of experimental hydrogen flux data.

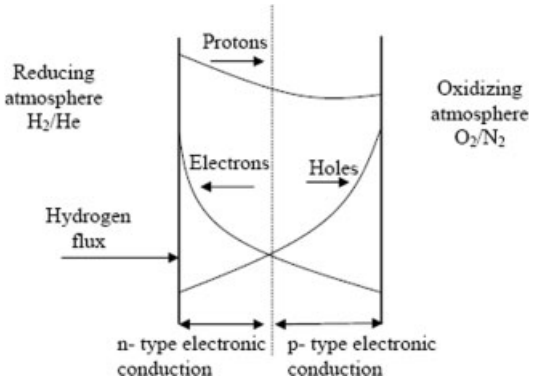


Figure 2.9 Charge carrier concentration profiles inside proton-conducting ceramic membranes.

Li and Iglesia [127] and Song et al. [124] numerically integrated Eq. (2.7). First, they determined the defect concentrations as a function of the O_2 and H_2 partial pressures within the membrane from the simulations based on a rigorous point defect model. Equilibrium constants for the defect reactions were obtained from the literature. Li and Iglesia [127] confirmed from their simulations that, under asymmetric conditions, the nature of the electronic conduction changes from n-type to p-type within the membrane as one moves from the reducing atmosphere side to the oxidizing atmosphere side. In the transition region, the transport number of electrons became very low. Tan et al. [128], in their development, did not neglect the oxide ion transport number and integrated Eq. (2.6) by a numerical method. However, they assumed that the electronic holes are the primary carriers, and neglected the transport number of electrons throughout the membrane thickness. This assumption is not justified, as the membrane is exposed to O_2 on one

side so the concentration of electrons (n-type) cannot be neglected in this case. Their results indicated that for Y-doped $SrCeO_3$, a transition from a protonic to an oxygen ionic conductor will occur at 1600 K.

2.5 Chemical Stability of Protonic Conductors

Two groups of perovskite-type ceramics, $BaCeO_3$ and $SrCeO_3$, offer high protonic conductivity due to their high basicity, which is advantageous for the formation of protonic charge carriers in a perovskite structure [129]. On the other hand, basic oxides are expected to react readily with acidic or even amphoteric gases such as SO_3, CO_2, or H_2O to form sulfates, carbonates or hydroxides. Hence, in spite of their tremendous potential, the use of these ceramics in practical processes has been limited due to their poor chemical stability under their respective operating conditions. In order to improve the stability of these protonic conducting perovskite-type oxides, it is important to investigate their chemical stability with respect to CO_2, H_2O or other gases, and to understand the poisoning mechanisms of these gases.

2.5.1 Stability in CO_2 Atmospheres

Most applications involve the presence of CO_2 from either air or oxidation products of organic fuels. Prolonged exposure to CO_2-containing atmospheres thus leads to mechanical disintegration of membranes as a result of the decomposition to carbonates and oxides. Many studies have been reported on the reaction of CO_2 with proton-conducting perovskite-type ceramics [129–138]. Thermodynamic data were obtained using various techniques such as galvanic cell [139], calorimetric [140, 142], thermogravimetric [135, 143], and tensimetric measurements [130]. The reaction temperature for $SrCeO_3$ in pure CO_2 is 917 °C and for $BaCeO_3$ it is reported to be 912 °C [136].

Several authors have pointed out that the enthalpy for the formation of perovskite-type oxides from the individual oxides exhibits a correlation with the perovskite tolerance factor, *t*, which is defined as:

$$t = \frac{(R_A + R_o)}{\sqrt{2}(R_B + R_O)} \tag{2.8}$$

Here, *R* is the radius of the ions, and the subscripts A, B and O represent the corresponding ions in ABO_3. This factor serves as a structure parameter to describe the extent of distortion of the perovskite structure from the ideal cubic prototype due to mismatch between the A–O and B–O bond lengths. Figure 2.10 shows the relationship of ΔH^0_{rxn}, the heat of formation of ABO_3 from oxide precursors, to the perovskite tolerance factor, *t*. As indicated in Fig. 2.10, the stability of the perovskite structure increases as the tolerance factor increases towards 1, because

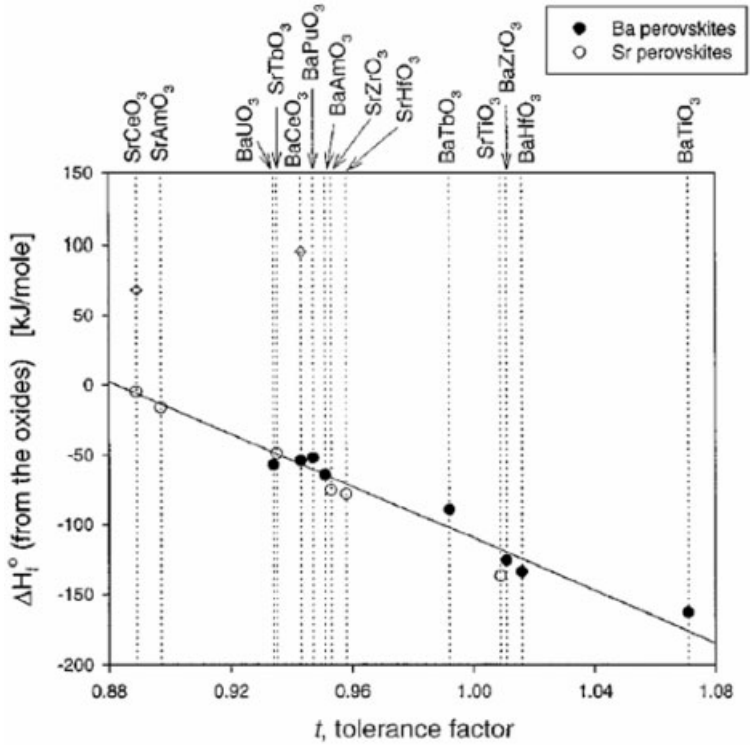


Figure 2.10 The enthalpy of formation of various perovskites from oxide precursors plotted as a function of perovskite tolerance factor [134].

the distortion from the cubic prototype decreases. For perovskite-type oxides with the same alkaline earth metals on the A-site, stability is mainly determined by the choice of the B-cation. In accordance with an increasing perovskite tolerance factor, one observes increasing stability in the order of: cerates < zirconates < titanates.

Haile et al. [134] studied the effect of different Zr doping levels on the stability of $BaCe_{0.90-x}Zr_xM_{0.10}O_{3-\delta}$ (M=Yb, Gd, Nd) against reaction with CO_2 using thermogravimetric analysis. The results of this study are summarized in Fig. 2.11. It was found that Gd-doped compounds containing 40 mol% Zr and Nd-doped compounds containing 20 mol% Zr did not react with CO_2 within the time frame of the experiment (about 1 h). It was also observed that the ceramic powder with higher calcination temperature was more stable in CO_2. This could be due to the greater chemical homogeneity of compounds calcined at higher temperatures. This also indicates that the chemical synthesis route is a preferable route for the synthesis of perovskite-type oxides because the dopant incorporation is more homogeneous as compared to the solid-state synthesis route.

The effect of Zr doping on the electrical conductivity has also been studied. The total conductivity of $BaCe_{0.90-x}Zr_xM_{0.10}O_{3-\delta}$ (M=Yb, Gd, Nd) in humid air decreases monotonically with increasing Zr content at low temperatures

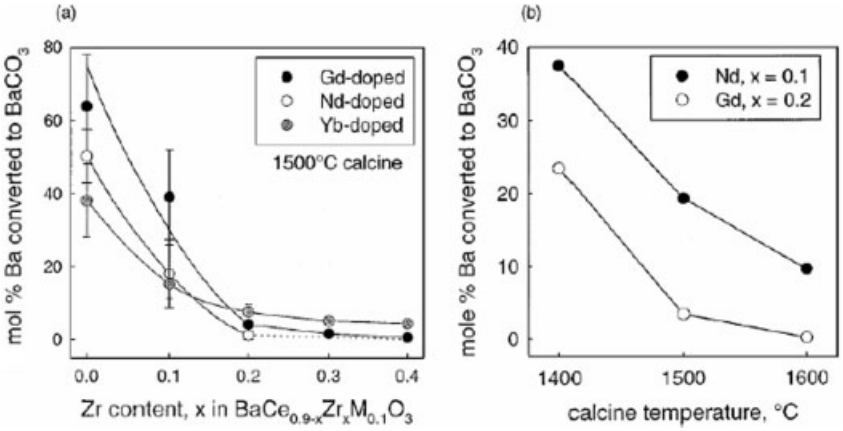


Figure 2.11 Maximum mol% of barium in $BaCe_{0.90-x}Zr_xM_{0.10}O_{3-\delta}$ converted to $BaCO_3$ during thermogravimetric analysis examination, as determined by the maximum weight gain upon heating [134].

(<400 °C). The impact of Zr substitution on proton conductivity increases in the order: Yb < Gd < Nd. In another study, it was found that the diminishing effect of Zr concentration on proton conductivity becomes less significant for $SrYb_{0.05}(Ce_{1-x}Zr_x)_{0.95}O_{3-\delta}$ at higher temperatures [144]. For $BaCe_{0.90-x}Zr_xNd_{0.10}O_{3-\delta}$, it was found that there was a very small decrease in the total conductivity in air for $x = 0$ to 0.6 in the temperature range 600–800 °C [145]. This observation suggests that $BaCe_{0.70}Zr_{0.20}Nd_{0.10}O_{3-\delta}$ should be a promising material with both high conductivity and chemical stability.

Besides doping an appropriate amount of Zr into cerates, other methods of modifying the composition have also been proposed to improve the stability for proton-conducting perovskite-type ceramics without seriously inhibiting their electrical conductivity [147, 148]. The choice of A-cation is almost independent of their stability with acidic gases. Barium is believed to be a suitable A-cation. The choice of B-cation, however, requires a compromise. It should be of medium size and should form no significant covalent bonds with its oxygen ligands. Small B-cations could reduce the water solubility, while large B-cations could result in poor compatibility and thus reduce the thermodynamic stability, including resistance towards poisoning by acidic gases. If B-cations are too acidic, the hydration enthalpy is not negative enough to retain the protonic defects up to the operational temperatures, whereas too basic B-cations could increase the proton transfer barrier beyond an acceptable limit. Normally, using a dopant on the B-site only (<30 mol%) is sufficient to match the acid/base properties to avoid defect association effects [136]. However, in practical applications, a 1:1 or 2:1 occupation of the B-site with cations of different acid/base properties is preferred because this situation seems to combine increased stability with small defect association effects.

2.5.2
Stability in Moisture-containing Atmospheres

Tanner and Virkar [148] found that pure and rare-earth doped $BaCeO_3$ were thermodynamically unstable in the temperature range 500–900 °C and decomposed to form CeO_2 and $Ba(OH)_2$. The endurance of perovskite-type oxides against water was examined by observing the appearance of the samples and the pH values of the solution [148, 149]. No changes in the dense pellets and powders were observed when exposed to water at 298 K. However, pellets disintegrated in the presence of some acid in water at room temperature and also in boiling water, due to dissolution of $BaCeO_3$ [148, 150]. The reaction mechanism includes dissolution of an alkaline-earth element such as barium from the perovskite-type solid into boiling water to form barium ions, and subsequent reaction of barium ions with carbon dioxide from the air to form barium carbonate. The reactions can be described as follows:

$$BaCeO_3 + nH_2O \rightarrow Ba^{2+} + 2OH^- + CeO_2 \cdot (n-1)H_2O \tag{2.9}$$

$$Ba^{2+} + 2OH^- + CO_2 \Leftrightarrow BaCO_3 + H_2O \tag{2.10}$$

It was found that the durability in steam for several perovskite-type oxides decreases in the order: $BaZrO_3 > BaZr_{0.5}Ce_{0.5}O_3 > SrZrO_3 > CaZrO_3 > SrCeO_3$, $BaCeO_3 > CaCeO_3$. It was concluded that stability against water vapor depends on the bonding strength of combination [149].

2.5.3
Stability in Reducing Atmospheres

In H_2 separation applications, the membranes are exposed to reducing atmospheres of H_2. Hence it is important to investigate the chemical and mechanical stability of the membranes in reducing atmospheres. Thermogravimetric analysis was performed for Nd-doped $BaCeO_3$ (BCNd) powder in H_2 [151]. A slight weight loss (~0.2%) was observed at temperatures higher than 400 °C which was caused by the desorption of lattice oxide ions from the material to form oxygen vacancies. When a sintered dark brown BCNd pellet was fired in H_2, it changed color to green but still demonstrated good mechanical strength. The color change may be caused by the partial reduction of Ce^{4+} to Ce^{3+} in the reducing atmosphere of H_2. In a recent study on Zr-doped $BaPrO_3$, it was found that disks became brittle after exposure to atmospheres of H_2, possibly due to their reduction in H_2 [152].

Kisho [153] evaluated the phase relations and oxygen stoichiometry of $SrCe_{0.95}Yb_{0.05}O_{3-\delta}$ under various partial pressures of O_2 and at different temperatures using X-ray diffraction and thermogravimetric analysis. He reported that there was no decomposition observed. However, he did not mention the partial pressures of H_2O in the purge gases. Okada et al. [154] found that the perovskite phase of SCYb decomposed to Sr_2CeO_4 under strongly reducing atmospheres

such as dry H_2 ($P_{O2} = 3.1 \times 10^{-15}$ Pa, $P_{H2O} = 3.1 \times 10$ Pa) at 1273 K. This was also confirmed from Raman spectroscopy. However, the specimen was stable in the case of a less reducing atmosphere of $P_{H2O} = 7.4 \times 10^{-3}$ Pa ($P_{O2} = 3.0 \times 10^{-14}$ Pa) at 1273 K. The origin of this decomposition is probably due to the reduction of Ce^{4+} to Ce^{3+}.

2.6 Future Directions and Perspectives

An ideal membrane must have high electronic and protonic conductivities; it should be fairly thin (1–10 μm), and chemically stable for prolonged periods of time under the presence of various gases such as CO_2, H_2O and H_2S etc. The low electronic conductivity limits the H_2 flux of the proton-conducting ceramic membranes. Under pure H_2 atmosphere at 900 °C, the protonic conductivities of SCYb and BCNd are about $0.7 \times l0^{-2}$ and $2.2 \times l0^{-2}$ S cm^{-1}, respectively. These values are slightly lower than the oxygen ionic conductivity of yttria stabilized zirconia (YSZ) or lanthanum strontium cobaltite (LSC) perovskite-type ceramics in air at the same temperature, so that there is room for further improvement in protonic conductivity.

$SrCeO_3$ and $BaCeO_3$ have high protonic conductivity due to the basic nature of the A- and B-site cations, but it makes them vulnerable to attack by acidic gas such as CO_2, SO_3 and H_2O. In the coming years research efforts will be continued to find an optimum material of the composition $ACe_{1-x-y}B_xM_yO_{3-\delta}$ for mixed electronic and protonic conductivity and high stability. Here M represents a lanthanide element, and B could be Zr or Ti as they provide stability towards CO_2 and other acidic gases. A molecular modeling approach, to study the effect of doping by aliovalent ions on the creation of oxygen vacancies and the distortion of the oxygen sub-lattice around the oxygen vacancies, offers promise in predicting the optimized material composition [155]. This distortion was recently suggested to enable the pathway for the proton diffusion between the second nearest neighbor O ions in perovskites.

Studies have revealed that much higher H_2 permeation flux can be obtained if the dense layer can be made as thin as 5 μm, or even less. This is because surface reaction is less likely to be a rate-limiting step for hydrogen permeation through mixed-conducting ceramic membranes as compared to oxygen permeation through similar types of membranes. Research efforts will be directed to find cost-effective ways to prepare such dense, thin films with techniques that are not complicated and have a high degree of reproducibility.

Although a variety of proton-conducting ceramic materials has been found in the past decade, only very limited studies have dealt with experimental measurements of H_2 flux, and little effort has been made to model the experimentally measured data in these studies. H_2 permeation flux data on more perovskite-based membranes are needed so that the mathematical models can be verified with experimental data. Such models, when rigorous and accurate, can provide essential guidance in the design and understanding of the mixed conducting materials. Fundamental study should be conducted to understand the structural,

chemical and electronic properties of the proton-conducting ceramic membranes under hydrogen permeation conditions.

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